

Proof of the Yang-Mills Gap Theorem

Chapter 2: Instantons and Tunneling in Yang-Mills Theory

2.1 Introduction to Instantons

Instantons are solutions to the Yang-Mills field equations in Euclidean space that represent tunneling events between different vacuum states. First introduced by Belavin, Polyakov, Schwarz, and Tyupkin in 1975, instantons have become a central tool in understanding non-perturbative effects in quantum field theory. Their importance lies in their ability to connect distinct vacua, thus providing a mechanism for phenomena such as quantum tunneling and the generation of a mass gap in Yang-Mills theory.

2.2 Mathematical Formulation

The instanton solution in Euclidean Yang-Mills theory is obtained by considering the field strength tensor $F_{\mu\nu}^a$ in the context of a gauge group $SU(2)$. The field equations in Euclidean space are:

$$D_\mu F^{\mu\nu} = 0$$

where D_μ is the covariant derivative. Instantons are self-dual solutions, satisfying:

$$F_{\mu\nu} = \tilde{F}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$$

This condition ensures that the Euclidean action S_E is minimized:

$$S_E = \frac{1}{2g^2} \int d^4x F_{\mu\nu}^a F^{\mu\nu a} = \frac{8\pi^2 n}{g^2}$$

where n is the topological charge, representing the winding number of the gauge field configuration.

2.3 Topological Charge and Energy Quantization

The topological charge n is an integer that classifies the instanton solutions according to how many times they wrap around the vacuum manifold. This

quantization leads to discrete energy levels associated with different instanton configurations:

$$E_{\text{instanton}} = \frac{8\pi^2 n}{g^2}$$

Since n is non-zero for non-trivial instanton configurations, the energy of the system is always positive, thereby preventing the existence of massless states. This quantization is key to establishing the mass gap in Yang-Mills theory.

2.4 Implications for the Mass Gap

The existence of a positive energy contribution from instantons directly implies a positive mass gap in the theory. As instantons enforce energy quantization, they ensure that the vacuum cannot support excitations with zero energy. Therefore, the smallest possible energy of a non-trivial state is strictly positive, confirming the existence of a mass gap.